

STEEL STRUCTURE CONVERSION

Steel 2D→3D Conversion (E3D)

General steel-structure conversion pipeline (00–03) generating beams / handrails / grating / columns as E3D SCTN/PANE.

Key Purpose

Why this program

1. One DXF pick builds the whole steel model

– Beams, handrails, grating and columns are all converted in a single Apply run

→ **Effect:** A 2D steel drawing becomes a ready E3D macro in one click instead of tracing each member by hand

2. Native E3D SCTN / PANE output

– Members export as SCTN, grating and plates as PANE, with an origin marker at NO E0

→ **Effect:** The macro drops straight into E3D and lands on the correct grid origin with no manual re-positioning

3. Drawing conventions read automatically

– COL NO grids, RED centre lines and TOS elevations are detected and turned into placement aids

→ **Effect:** Members sit at the right position and height, so downstream stress/clash work starts from a trustworthy model

4. One combined macro per job

– All part macros are merged into a single .all.mac and the Open button launches it in the editor

Required inputs : One or more Steel Structure 2D DXF files (with COL NO grid, RED centre lines, TOS elevations)

Main Window – Item Guide

The screenshot shows the 'Steel Structure Modeling Option' dialog box. It has a title bar with a gear icon and the text 'Steel Structure Modeling Option'. Below the title bar is a tab labeled 'General'. The main area is titled 'Select Steel Structure 2D dxf File :'. It contains a large empty list box (1) with the text 'No file selected' below it. Below the list box are four buttons: a blue button with a plus icon (2), a 'Remove Selected' button (3), a 'Clear All' button, and a disabled button with a minus icon (4). Below these are two sections: 'Aid Steel Size Expression' with buttons for 'Symbol' (5), 'Size', and 'Off'; and 'Hand Rail Level' with buttons for '2 Level' (6) and '3 Level'. Below these is the 'Aid Expression' section with three checked checkboxes: 'Col No', 'Red' (7), and 'Section'. Below the checkboxes are four input fields: 'Hand Rail Height : 1232' (8), 'Macro File Create : (DXF 도면과 동일)' (9), 'Stru Ori. North/East : 0' (10), and 'Location Plan Column : North(V):East' (11). At the bottom are four buttons: 'Apply', 'Save' (12), 'Load', and 'Cancel'.

1. Selected 2D DXF file list
2. Select DXF files
3. Remove Selected / Clear All
4. Open generated macro in editor
5. Aid Steel Size: Symbol / Real Size / Off
6. Hand Rail Level: 2 or 3
7. Aid Expression: Col No / Red Center / Section
8. Hand Rail Height (mm)
9. Macro File output folder
10. Stru Origin North / East offset
11. Location Plan Column mapping
12. Apply / Save / Load / Cancel

Processing Steps

What 'Apply' runs, in order

1. Beam → SCTN + AID overlay
2. Handrail → STRU / SUBS / BOX / CYLINDER
3. Grating → PANE
4. Column → vertical SCTN
5. Merge part macros into one _all.mac
6. Insert NO E0 origin marker and enable Open

Overview

What it does

1. General steel-structure conversion pipeline (00–03) generating beams
2. Handrails
3. Grating
4. Columns as E3D SCTN/PANE

Category: Steel Structure Conversion

Folder: STEEL_STRUCTURE/2DTO3DFORE3D

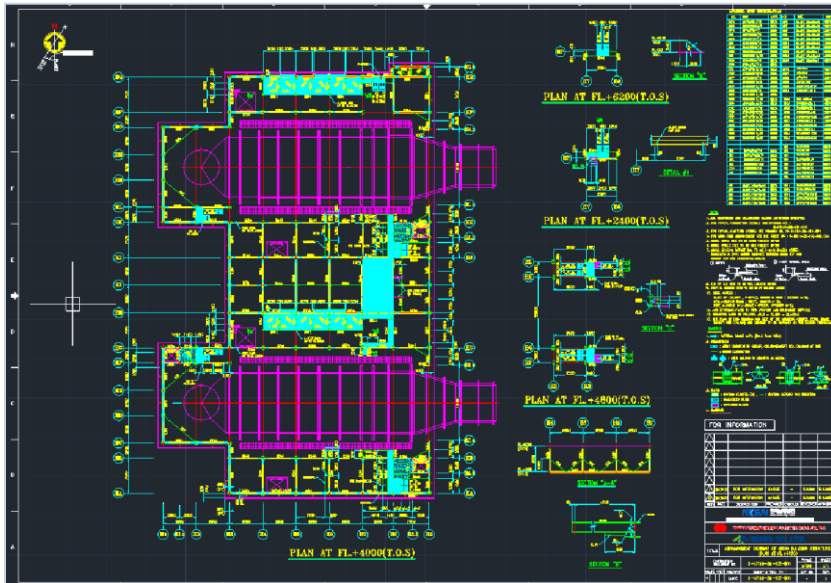
Main scripts: ace_steel_runall.PY

Scripts / Status: 13 scripts · Operational

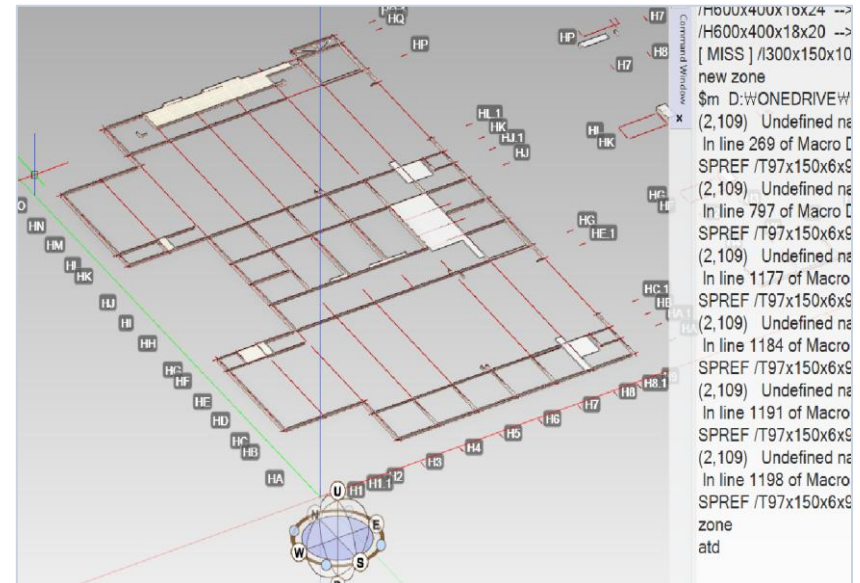
Tags: E3D · SCTN · PANE · Steel

Output Samples

Output Samples (1/4)



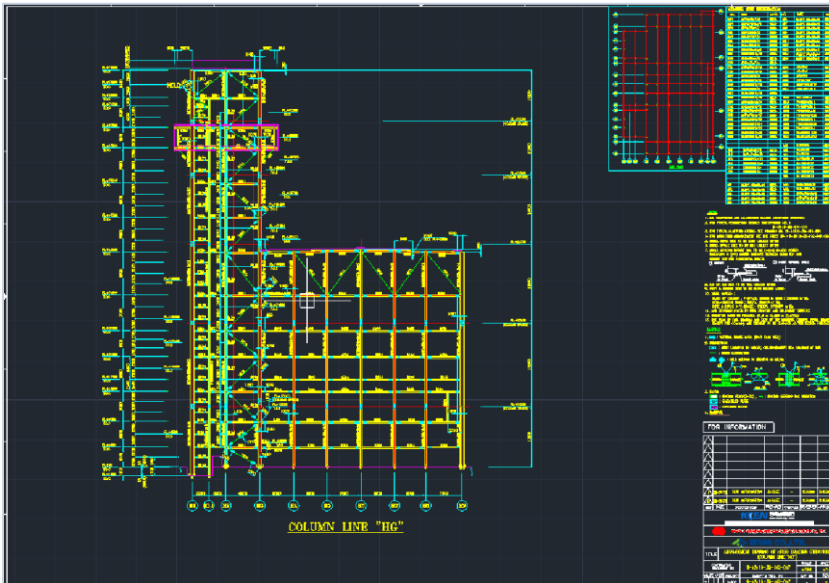
input1-1



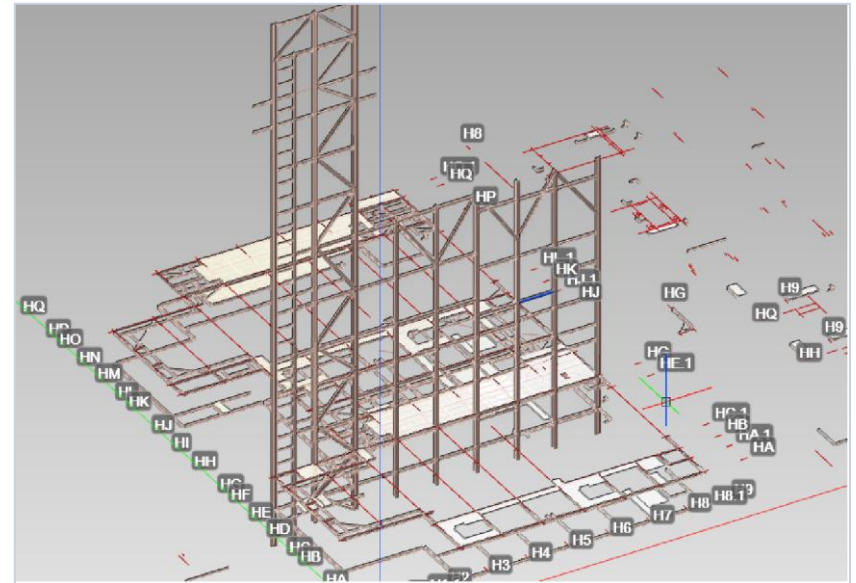
output1-1

Output Samples

Output Samples (2/4)

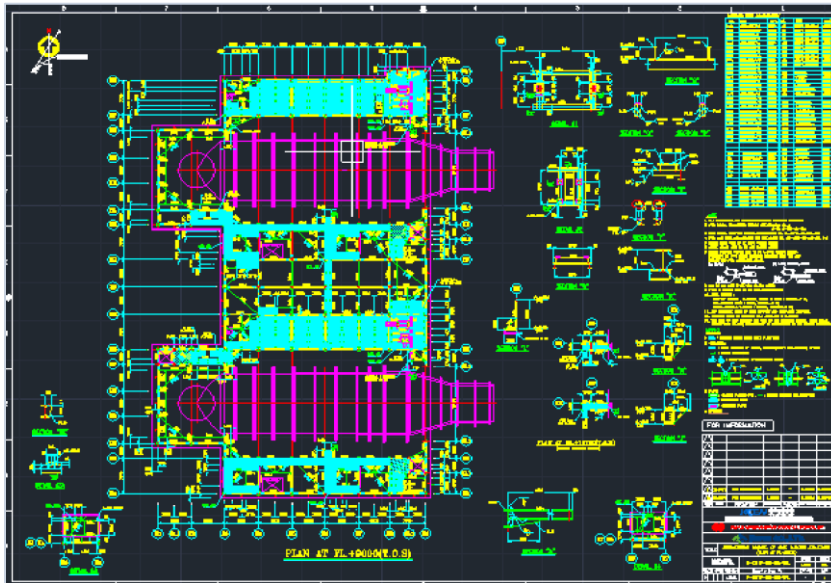


input1-2

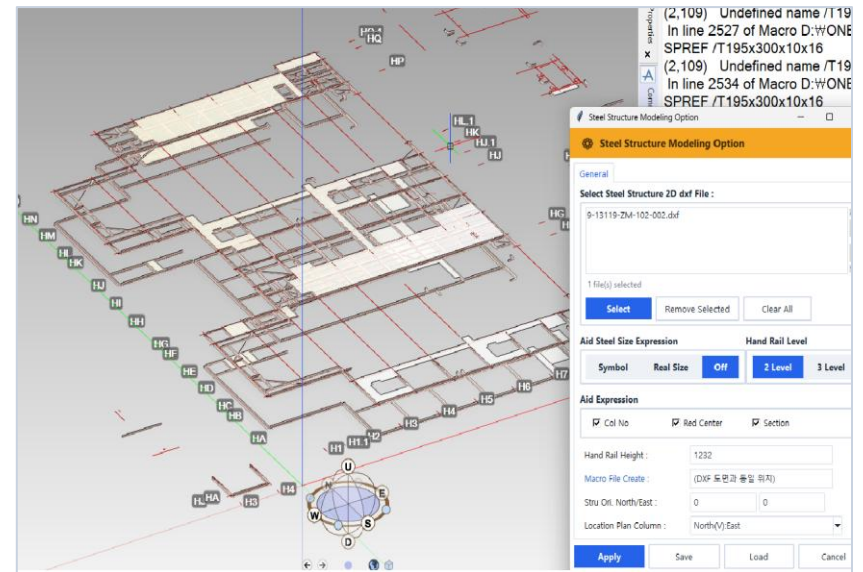


Output Samples

Output Samples (3/4)

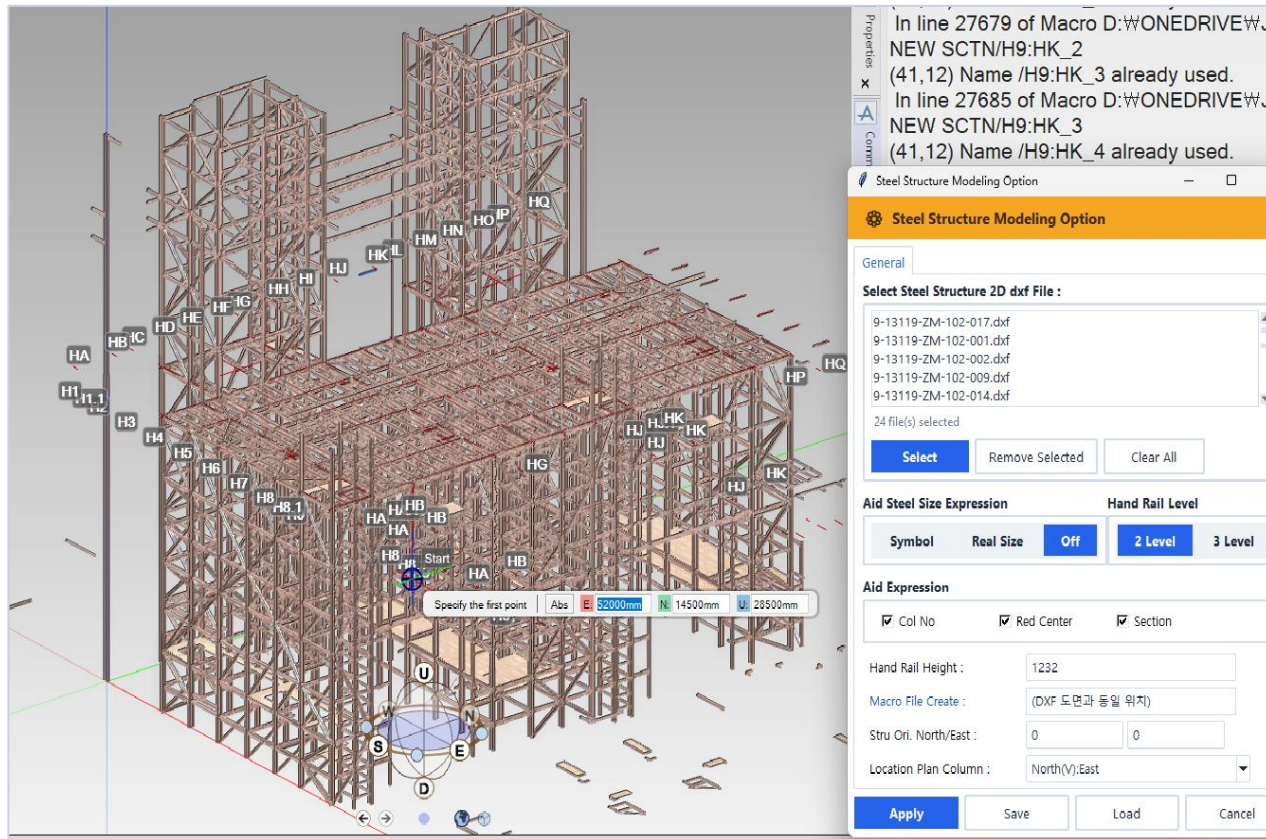


input1-3



output1-3

Output Samples (4/4)



outputall

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Steel Structure Conversion.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: ace_steel_runall.PY.
- Output samples and the program window above reflect the current build.

Thank You

Steel 2D→3D Conversion (E3D) · Plant Engineering Total Service System